

Ali Azhar



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PROFESSIONAL EXPERIENCE

Associate AI/ML Engineer, AMIIQ

09/2024 – present | Lahore, Pakistan

- Led **AI** research and developed **Deep Learning** solutions in **Computer Vision** for medical imaging and **Natural Language Processing** for clinical text analysis, enhancing diagnostic accuracy.
- Built an **NLP-driven** X-ray report summarization system, integrating multiple medical APIs with **FastAPI** for efficient processing and **Docker** for scalable deployment.
- Trained a **Biomedical BERT** model for **Named Entity Recognition (NER)**, achieving a **92% F1-score** in extracting key features from X-ray reports using **Azure ML Studio**.
- Developed **Computer Vision** models for fetal part analysis, including:
 - Semantic segmentation** (U-Net, AF-Net, nnU-Net): **97% DICE, 91% IOU**
 - Object detection** (YOLO-based): **92.5% mAP**
 - Classification** (EfficientNet, MobileNet): **94% accuracy**
- Designed standardized pipelines for data preprocessing, model training, and evaluation.
- Deployed AI solutions on **Azure**, streamlining cloud-based data ingestion and transformation.
- Maintained version-controlled documentation on **GitHub** for reproducibility and compliance.

IT Specialist, 786 Outsource Ltd

07/2023 – 08/2024 | Lahore, Pakistan

- Developed **ML** models and predictive analytics to enhance decision-making, integrating structured and unstructured data sources.
- Designed and maintained data-driven applications using **Java**, optimizing backend logic, database performance, and system efficiency with **SQLite**.
- Built interactive dashboards in **Power BI** to track inventory system performance and business analytics.



SKILLS

- Languages: Python, Java, C++, SQL, JavaScript, HTML, CSS
- Tools and Frameworks: PyTorch, TensorFlow, Power BI, OpenCV, Scikit-learn, Pandas, NumPy, Matplotlib, Tkinter, NLTK.
- IDEs & Development Platforms: Visual Studio Code, Spring Boot, NetBeans
- Operating Systems & Environments: Linux (Ubuntu), Windows, CPU/GPU Computing
- Others: Azure, Git, Jira, Docker, SQLite, FastAPI



EDUCATION

Bachelor of Science in Data Science, FAST NUCES

09/2020 – 08/2024 | Lahore, Pakistan

A - LEVELS, Beaconhouse College Campus Gulberg

09/2017 – 06/2019 | Lahore, Pakistan



AWARDS

Dean's List of Honor, FAST NUCES

2022

Bronze Medal Holder, FAST NUCES

2022



PROJECTS

Chest X-ray Report Summarization System

02/2025 – present

Developed an **NLP-driven** X-ray report summarization system that takes a chest X-ray as input, generates reports from multiple medical APIs, and summarizes them. Fine-tuned a **Biomedical BERT** model on a self-labeled dataset for **NER** to extract key features, then calculated similarity between reports based on their text and findings, leveraging **FastAPI** for integration and **Docker** for scalable deployment.

Automated Fetal Gestational Age (GA) Estimation

11/2024 – 01/2025

Developed and deployed on **Azure** an automated system for fetal GA estimation using a **ResNet**-based classification model to filter fetal structures from garbage data, followed by **V-Net** segmentation and a CNN-based feature extraction model to segment key fetal structures, achieving a **93% DICE score** and enabling accurate GA calculation from segmented areas.

Stressful Emotion Detection and Classification Using Audio Data

09/2023 – 05/2024

Developed an end-to-end audio-based emotion recognition system, leveraging **VGGish** architecture to achieve **86% accuracy** in detecting stress and from voice patterns. Engineered a comprehensive dataset of **1,000** diverse audio samples from young adults (15-25), utilizing **Librosa** for advanced acoustic feature extraction including **MFCCs**, spectral contrast, and chromagrams.